md-to-pdf

A web service for converting Markdown to PDF

Web UI

For quick experimentation, you can use the web version hosted on Fly.io. Paste your Markdown and download the converted PDF.

Availability of the service is not guaranteed, see Fly.io status when it is down. If you need guaranteed availability, deploy it yourself.

API

You can convert Markdown by sending a `POST` request to `https://md-to-pdf.fly.dev`.

````
curl --data-urlencode 'markdown=# Heading 1' --output md-to-pdf.pdf https://md-to-pdf.fly.dev
```B.Tech CSE 1st Year
Complete Academic Guide 2025

Prepared by: Grok — Built by xAI
Version: January 2025
For: First Year Computer Science Engineering Students

Table of Contents
Introduction
Semester 1: Subjects & Detailed Overview
Semester 2: Subjects & Detailed Overview
Most Important Topics for CSE Students
Recommended Books & Resources
Study Strategy & Pro Tips
Common Mistakes to Avoid
1st Year Roadmap & Future Preparation
Introduction
Welcome to B.Tech in Computer Science and Engineering.

The first year is the foundation year. Your performance here
decides how strong your basics will be for the next three years.
This guide covers everything you need — syllabus overview,
important topics, best resources, and smart study strategies.

Key Focus for CSE Students:

Master C Programming thoroughly
Build strong mathematical foundation
Develop problem-solving skills
````
````

Understand basic electronics and physics

Semester 1: Subjects & Overview

Subject Name	Credits	Key Topics	Weightage
BSC-101 Mathematics – I	4	Calculus, Matrices, Sequences, Series	High
BSC-102 Engineering Physics	4	Optics, Quantum Mechanics, Semiconductors, Lasers	Medium
ESC-101 Basic Electrical Engineering	3	DC & AC Circuits, Transformers, Motors	Medium
ESC-102 Programming for Problem Solving (C)	4	C Fundamentals, Arrays, Pointers, Structures, File Handling	Very High
HSMC-101 English & Communication Skills	2	Technical Writing, Grammar, Presentations	Low

Practicals:

Physics Lab
Programming Lab (C Language)
Workshop Practices

Semester 2: Subjects & Overview

Subject Name	Credits	Key Topics	Weightage
BSC-201 Mathematics – II	4	Probability, Statistics, Complex Variables, Laplace	High
BSC-202 Engineering Chemistry	4	Polymers, Fuels, Corrosion, Water Chemistry	Medium
ESC-201 Basic Electronics Engineering	3	Diodes, Transistors, Op-Amps, Digital Logic	High
ESC-202 Data Structures & Algorithms	4	Arrays, Linked Lists, Stacks, Queues, Trees, Sorting & Searching	Very High
ESC-203 Engineering Graphics & Design	3	Orthographic & Isometric Projections, AutoCAD	Medium

Practicals:

Chemistry Lab
Data Structures Lab
Electronics Lab
Computer Aided Design Lab

Most Important Topics for 1st Year CSE

Programming (C Language):

Pointers & Dynamic Memory Allocation
Arrays and Strings
Functions and Recursion
File Handling
Structures and Unions

Mathematics:

Differential & Integral Calculus
Linear Algebra (Matrices & Determinants)
Probability & Statistics

Data Structures:

Linked Lists, Stacks, Queues

Binary Trees
Sorting Algorithms (Quick, Merge, Heap)
Electronics & Physics:

Semiconductor Physics
Digital Logic Gates
Transistors and Op-Amps
Recommended Books & Resources
Best Books:

Let Us C – Yashavant Kanetkar
The C Programming Language – Kernighan & Ritchie
Engineering Mathematics – B.S. Grewal
Data Structures Using C – Reema Thareja
Basic Electronics – Boylestad & Nashelsky
Best YouTube Channels:

Neso Academy
Gate Smashers
Physics Wallah (Mathematics)
Last Moment Tuitions
CodeWithHarry (C Programming)
Online Platforms:

GeeksforGeeks
Programiz
NPTEL
LeetCode (for practice)
Smart Study Strategy for 1st Year
Daily Coding Practice – Minimum 1 hour of coding every day.
Mathematics is compulsory – Never skip it. It helps in almost all future CSE subjects.
Maintain regular notes and revise weekly.
Complete all lab assignments on time and maintain neat lab records.
Aim for CGPA 8.0 or above – It helps in placements and higher studies.
Start learning Python and Git in your free time.
Weekly Time Allocation Suggestion:

Programming + DSA: 15 hours
Mathematics: 12 hours
Physics/Electronics/Chemistry: 10 hours
English + Soft Skills: 3 hours
Common Mistakes to Avoid
Treating Programming as a theory subject
Ignoring practicals and viva preparation
Last-minute preparation before exams
Not practicing numerical problems in Mathematics
Underestimating Data Structures in 2nd semester
Final Tips
Your 1st year performance sets the tone for your entire B.Tech

````
journey. Focus on conceptual clarity rather than mugging up.
Build strong fundamentals in C, Mathematics, and Data Structures
– these will help you throughout your career.

Pro Tip:
Start solving previous year question papers from your university
(AKTU, VTU, JNTU, MAKAUT, etc.) at least one month before exams.



| Parameter  | Required |
Description
|
|------------|----------|--------------------------------------------------
| `markdown` | Required | The markdown content to
convert
|
| `css`      | Optional | CSS styles to
apply
|
| `engine`   | Optional | The PDF conversion engine, can be
`weasyprint`, `wkhtmltopdf` or `pdflatex`, defaults to
`weasyprint` |

Send data from files like this:

```shell
curl --data-urlencode "markdown=$(cat example.md)"
````

Deploy

A prebuilt container image is available at [Docker Hub](). The container starts up the web service and listens for HTTP on port 8000.

You can run it yourself like this:

```
docker run --publish=8000:8000 spawnia/md-to-pdf
```

You may configure the webserver through [Rocket environment variables](). For example, you could allow larger payloads by increasing the limit for form data:

```
ROCKET_LIMITS={form="1MiB"}
```

Built with

- Rocket - a web framework for Rust
- Pandoc - a universal document converter
- Codemirror - a text editor for the browser
```